Jan 22

★ Announcements:

- Course schedule updated, HW due Thursdays
- SMIMIC events, free pizza, talks ~every other Thursday
  - Feb 5th SMIMIC may be pushed back/rescheduled

HW 0

- Notation: $\mathbb{Z}/m\mathbb{Z} = \mathbb{Z} \bmod m\mathbb{Z}$
  $= \mathbb{Z}/m = \mathbb{Z}/\langle m \rangle = \mathbb{Z}_m$

7) $x^2+1=0$ no solutions in $\mathbb{Z}$ (or even in $\mathbb{R}$)

Modulo 2:

$1^2+1=2\equiv 0 \bmod 2$, $3^2+1=10\equiv 0 \bmod 2$ So mod 2 works

Plug in any odd #, $(2n+1)^2+1=4n^2+4n+1+1\equiv 0 \bmod 2$

- We partition the integers into equivalence classes
  congruence, residue, etc

$[0]_2 = \{n\in\mathbb{Z} : n\equiv 0 \bmod 2\}$
$= \{n\in\mathbb{Z} : 2\mid n\}$ divides

Zero is used as a representative, $[0]_2=[2]_2=[4]_2=[-2784\,18]_2$, etc
when context is clear, we use the bar $\overline{0}=\overline{2}=\overline{4}=\overline{\text{even number}}$

★ note: the bar notation depends on context, not always mod 2

- Q: congruence $\equiv$ vs. equality $=$

Q.9 shows $\equiv$ is an equivalence relation, so they behave similarly (not the same)

- Use congruence when relating integers under/via a modulus (things are similar in mod m)
- Use equality when talking about equivalence classes in $\mathbb{Z}/m\mathbb{Z}$ (things are the same)

ex. $16\equiv -1 \bmod 17$, but $16\neq -1$

$[16]_{17}=[-1]_{17}$ in $\mathbb{Z}/m\mathbb{Z}$
$= \{n\in\mathbb{Z} : n\equiv 16 \bmod 17\} \longrightarrow \{n\in\mathbb{Z} : n\equiv -1 \bmod 17\}$

- We don't say $[16]_{17}\equiv[-1]_{17}$, congruence relates integers, not equivalence classes
- $[x]_m^2+[1]_m=[0]_m \longrightarrow$ you can do the operations on the representatives
  same as $[x^2]_m+[1]_m$, same as $[x^2+1]_m$

Ch. 2: Groups

- Def Let G be a set, and let $*: G\times G \longrightarrow G$ be a well-defined binary operation. We say G is a group if the following are satisfied:

1) for any $a,b,c\in G$, $(a*b)*c = a*(b*c)$ (associativity)

2) There exists $e \in G$ (id, 1, etc) such that $e*a = a*e = a$ for all $a \in G$ (there exists an identity for $*$)

3) for every $a \in G$ there exists an inverse in G such that $a*a^{-1} = a^{-1}*a = e$ (every element in G has an inverse)

Implicit condition from $*: G \times G \to \boxed{G}$ is closure

ex. $\mathbb{N}$ is closed under $+$, but not under $-$ (ex. $2-4$ undefined)

- If, additionally, G is commutative under $*$, we say G is Abelian (an Abelian group)
  $a*b = b*a \ \forall a,b \in G$
  - Subtraction is not commutative, associativity also fails (often) for subtraction
  - Matrix multiplication is not commutative
- Examples of groups:
  - $\mathbb{Z}$ under addition is an Abelian group abbreviated $(\mathbb{Z},+)$
    - Closed under $+$, ex. $7+(-35) = -28$ ✓
    - Associative under $+$, ex. $(1+2)+3 = 1+(2+3) = 6$ ✓
      ($(1+2)+3 \to 3+3$, $1+(2+3) \to 1+5$)
    - Identity under $+$ (namely 0), ex. $5+0 = 5 = 0+5$ ✓
    - Inverses under $+$ (negative/positive), ex. $12+(-12) = 0 = (-12)+12$ ✓
    - Abelian: Commutative under $+$, ex. $3+5 = 8 = 5+3$ ✓
  - $(\mathbb{Q},+)$, $(\mathbb{R},+)$, $(\mathbb{C},+)$, $(\mathbb{Z}/2\mathbb{Z},+)$, $(\mathbb{Z}/m\mathbb{Z})$, (2×2 matrices with coefficients in $\mathbb{R}$, $+$) are Abelian groups
- $(\mathbb{Z},\cdot)$ (mult.) is not a group (inverses, i.e 2 does not have an inverse)
- $(\mathbb{Q},\cdot)$ is not a group (no mult. inverse for zero)
  - $(\mathbb{Q}-\{0\},\cdot)$ is an Abelian group, $(\mathbb{Q}^+,\cdot)$ (positive rationals) is an Abelian group
    notation
    $\mathbb{Q}^\times = \mathbb{Q}^* = U(\mathbb{Q})$ = elements of $\mathbb{Q}$ with mult. inverses
  - $(\mathbb{R}^\times = \mathbb{R}-\{0\},\cdot)$ is an Abelian group

- $\mathbb{Z}^\times = \{n \in \mathbb{Z} : \exists n^{-1} \in \mathbb{Z} \text{ with } n \cdot n^{-1} = 1\}$
  $= \{1,-1\} = \{\pm 1\}$
  - Prove directly that $(\{\pm 1\},\cdot)$ is an Abelian group

$\cdot$	1	-1
1	1	-1
-1	-1	1

Inverses ✓ (1 in every row) assoc. ✓ (takes some more argument)
Identity ✓ (unchanged elements)
Commut. ✓ (row·column = column·row, table reflects across diagonal ↘)